MIL-S-45004E
27 April 1987
SUPERSEDING
MIL-S-45004D
22 July 1974

MILITARY SPECIFICATION

STAND, MAINTENANCE, AUTOMOTIVE ENGINE

(6,000 POUND CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a universal engine stand for manual handling, holding, and conveying V-type, in-line, and opposed type automotive engines weighing up to 6,000 pounds (see 6.1).

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-13-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Box, Wood, Nailed and Lock Corner
PPP-B-636 - Box, Shipping, Fiberboard
PPP-C-843 - Cushioning Material, Cellulosic
P?P-T-60 - Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-52950	Crate, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards For Federal Services

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910)

(Copies of specifications, standards, handbooks, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The automotive engine maintenance stand shall be new and meeting the performance requirements specified herein. The engine stand shall be designed for both mobile and stationary use while supporting engine assemblies during the performance of maintenance and repair work involving the disassembly and reassembly of the engine components. All engine stand parts subject to wear, breakage or distortion shall be accessible for adjustment and repair. All operating controls shall be so designed and positioned that they are visible and accessible to the operator and shall be designed so that operator's hands will not be subjected to pinch points, sharp edges, or snagging hazards.

3.2.1 Maintainability. Item configuration and adjustment features shall be designed to reduce maintenance operations to the simplest procedures compatible with a minimum of maintenance skills and standard maintenance equipment. Instructions for disassembly, repair, adjustment and reassembly shall be concise and clearly defined.

3.3 Construction. All parts and materials used in fabrication of the engine stand shall be free of defects. All surfaces of all parts shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. However, such parts subject to adjustment or removal shall not be permanently deformed.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with metallic pastes be used for reclaiming defective castings or forgings.

3.3.2 Welding. Welding shall be of a quality which shall sustain all requirements of the welded parts. These operations shall not be employed as repair measures for defective parts.

3.3.3 Safety provisions. Safety devices shall be provided for all parts presenting safety hazards which can be guarded. Guards shall not interfere with the operation of the stand. All guards shall be removable to provide access to guarded parts. Except as otherwise specified by the procuring activity, the stand shall comply with those standards promulgated under OSHA 2206 applicable to the stands themselves. Exceptions and additional requirements for safety shall be as specified (see 6.2 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

3.3.4 Lubrication. All bearings (except sealed-for-life type), mating gears, and other working parts shall be provided with standard means for lubrication. All oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.3.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.6 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.3.7 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the automotive equipment industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Performance and product characteristics.

3.4.1 Capacity. The stand shall be capable of safely accommodating, holding and conveying both in-line, V-type, and opposed type engine assemblies up to, and including, 6,000 pounds. The maximum dimensions for the in-line engine are 59 inches long, 39 inches wide and 51 inches high. The maximum dimensions for the V-type are 58 inches long, 39 inches wide and 55 inches high. The maximum dimensions for the opposed type engine are 42 inches long, 52 inches wide and 35 inches high. The engine stand when supporting any engine assembly within these parameters, shall be capable of being moved over a 1 inch 15 degree rise in the floor and a subsequent 1 inch vertical drop to the floor without tipping over or being structurally damaged or losing the load. The engine stand when supporting any engine assembly within these parameters, shall be capable of being pushed at a rate of not less than 1 foot per second against a 2 inch high vertical rise 90 degrees to the direction of movement at an approach angle which will bring two wheels or casters into contact with the rise at the point of greatest instability. The stand shall not lose the load, tip over, or sustain functional damage.

3.4.2 Engine assembly accessibility. The engine assembly components shall be accessible with no part of the engine stand interfering with the dismantling, correction, or reassembly of the engine components. The engine assembly components shall consist of the items and accessories from the bell housing forward.

3.4.3 Holding positions. The stand shall permit rotation of the engine assembly 360 degrees in either direction around a horizontal axis. The center line of rotation shall not be less than 36 inches from the floor (see 3,5.4). The stand, when rotation is stopped, shall automatically lock in any position along the 360 degrees without any operator action.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

3.4.4 Overload capacity. The engine stand and all components shall be capable of withstanding, without damage or deformation, an overload of 150 percent of the rated capacity (9,000 pounds) for a period of not less than 30 minutes.

3.4.5 Tilt and lift capability. The engine stand shall have a rotational axis tilt and lift capability in order to accommodate rotation of unusually long or wide engine assemblies (see 3.6).

3.5 Major components.

3.5.1 Frame. The engine stand frame shall consist of a base and as a minimum two upright posts or columns. The frame shall be fabricated of structural steel, tubular stock, or other material suitable to meet the requirements specified herein. The base shall have provisions for mounting the wheels and casters specified in 3.5.2. One of the upright posts or columns shall be movable to accommodate loading the engine assemblies to and from the stand. The movable post or column shall be provided with a positive means for rigidly locking it in position. The permanent post or column shall be designed to accommodate the necessary mechanism for manual rotation of engine assemblies. The opposite end of the supporting column or post shall be both removable and telescoping to allow fastening to an engine assembly or bell housing when required. The automotive engine maintenance stand shall accommodate the engines specified in Table I.

TABLE I. Engines Managed By TACOM

I/N	Manufacturer	Model Number	Drawing/Part No.	Federal Supply Code of Manufacturer (FSCM)
1	Allis-Chalmers	1700 Mark II	4392154	09367
2	Allis-Chalmers	6MB230	T1P0091-05 Nov 70	97403
3	Allis-Chalmers	6MB230	T1P01244-23 Apr 71	97403
4	Brunswick	A042	AD42	97403
5	Caterpillar	3238	3203APPL786	11081
6	Caterpillar	3306	4R8600	11033
7	Caterpillar	3306	5R3383	11083
8	Caterpillar	3306	5R4300	1 1083
9	Caterpillar	3306	7N4980	11083
10	Caterpillar	3408T	5R4553	11083
11	Caterpillar	1)333 TA	3R8704	11083
12	Caterpillar	D333	3R6103	11083
13	Caterpillar	D333	5R698	11083
14	Caterpillar	D333	5R829	11083
15	Caterpillar	D333C-T	62202	16004
16	Caterpillar	D333C	11681062	19207
17	Caterpillar	D333C	5S3817	11083
18	Caterpillar	D339	3R3012	11083
19	Caterpillar	D342	4N4445	11083
20	Caterpillar	D343	3R4222	11083
21	Caterpillar	D343	5R1514	11083
22	Chrysler	440 cu. in.		
23	Chrysler	LA 318-1 cu. in.	4219101	86403
24	Continental		FS244-6091	14351
25	Continental		M600A51017S	14351
26	Continental		SD802-2286	14351
27	Continental		SD80212304	14351

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

Engines Managed by TACOM - Continued

I/N	Manufacturer	Model Number	Drawing/PT No.	FSCM
28	Continental	F163		
29	Continental	F163	8610425	30612
30	Continental	F163	88402	78640
31	Continental	F163	F145A5033S	14351
32	Continental	F163	F162A5112S	14351
33	Continental	F163	F162A5113S	14351
34	Continental	F163	F400A5285S	14351
35	Continental	F163	F400A530701S	14351
36	Continental	F244	TIP0067-23 Mar 70	97403
37	Continental	F245	4871756	30612
38	Continental	FS162-6088	F145A5012AS	14351
39	Continental	FS244-6062	F244A50614S	14351
40	Continental	JD403	47122	14351
41	Continental	LD 465-1C	11641982	19207
42	Continental	LDS 427-2	11641710	19207
43	Continental	LDS 465-1A	11610187	19207
44	Continental	LDS 465-2	116101102	
45	Continental	LDT 465-1C	11642105	19207
46	Continental	R-6602	5703526	19207
47	Continental	Y112-861		
48	Continental	Y12		
49	Continental	YS116		
50	Cummins		99046-82	15434
51	Cummins	JN-6-1	99045-94	15434
52	Cummins	JN61P	99045-43	15434
53	Cummins	NHC 250	11664451	19207
54	Cummins	NHC 250	5704507	19207
55	Cummins	NT-1710-B525		
56	Cummins	NT380	99043-02	15434
57	Cummins	NTA-855-C-420		
58	Cummins	NTC 400 Big Cam I	MA87-21005	34623
59	Cummins	NTC 400 Big Cam II	5995180	34623
60	Cummins	NTF-365		
61	Cummins	V-903-C-265		
62	Cummins	V8-265	99045-44	15434
63	Cummins	V8-265BL	99047-30	15434
64	Cummins	V8-300	10946107-1	19207
65	Cummins	V903	99048-86	15434
66	Cummins	VTA-903T	8750013	15434
67	Detroit Diesel	1067-7040	1067-7040/S01D	
68	Detroit Diesel	12V71(T)	7123-7396	72582
69	Detroit Diesel	3-53 Series	11601645	19207
70	Detroit Diesel	3-53	5196490	72582
71	Detroit Diesel	3172	3172FWD144603	72582
72	Detroit Diesel	4-53	5043-7200	72582
73	Detroit Diesel	4-71	1043-7100	72582
74	Detroit Diesel	6.2 liter(HMMWV)	AMG5574076	72582
75	Detroit Diesel	6.2 liter(cucvs)	14067734	72582
76	Detroit Diesel	6V53	33001	13841
77	Detroit Diesel	6V53	5063-5000	72582

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

Engines Managed by TACOM - Continued

I/N	Manufacturer	Model Number	Drawing/PT No.	FSCM
78	Detroit Diesel	6V53	5063-5099	72582
79	Detroit Diesel	6V53	50635099	72582
80	Detroit Diesel	6V53N	510-282	27315
81	Detroit Diesel	6V53T 5063-5395	5704843	19207
82	Detroit Diesel	6V53T Alum Block	5702927	19207
83	Detroit Diesel	6V53T	5107220	72582
84	Detroit Diesel	6V71-S70	7063-7200RC	72582
85	Detroit Diesel	8V-71N	7083-7000	72582
86	Detroit Diesel	3V71T	5702938	19207
87	Detroit Diesel	8V71T	5703768	19207
88	Detroit Diesel	8V71T	5703769	19207
89	Detroit Diesel	8V71T	5704851	19207
90	Detroit Diesel	8V71T	5705020	19207
91	Detroit Diesel	8V71T	5705025	19207
92	Detroit Diesel	8V92 TA	1319410U	45152
93	Detroit Diesel	8V92T	8087-7340	72582
94	Detroit Diesel	8V92TA	8087-7440	72582
95	Detroit Diesel		1063-7100	72582
96	Detroit Diesel		4045C	72582
97	Detroit Diesel		5033-7001	72582
98	Detroit Diesel		5043-5198	72582
99	Detroit Diesel		5043-7201	72582
100	Detroit Diesel		5196490	72582
101	Detroit Diesel	N-53N	5043-7000	72582
102	Eaton	IND30	050080-00	83299
103	Fiat-Allis	3500 Mark II	74036089	90940
104	Ford		BAJA9931	23040
105	Ford		C5PF6002CF	90927
106	Hercules	L-141	11660530	19207
107	IHC	D-190	903611R94	31007
108	IHC	DT-466B	347693R91	29510
109	IHC	DT-817-C	684466C-92	92679
110	IHC	U-450	U-450	31007
111	IHC	UV-549	UV-549	31007
112	J. I. Case	207D	A154661	10988
113	J. I. Case		D15827	10988
114	LeRoi	T-H844		
115	M.A.N. VIO	D2840		
116	Mack	ENDT-673	5704120	19207
117	Mercedes Benz	617-952	4537800-801	53465
118	Reo	OA-331	7538638	19207
119	Waukesha	135GKC	135GKC	64104
120	Waukesha		195-GLC-5B	64104
121	Waukesha	F817G	F817G	64104
122	Waukesha	FC	FCBG26033	64104
123	Waukesha	FC	FCBG5943	64104
124	White	3400	D34X289	28265
125	White	G2300X244	13222E9851	97403
126	White	QXLDERQERX17		28265
127	White	QXLDERSPECQERX26		28265

Source: https://assist.dla.mil -- Downloaded: 202640-22T02:12Z
Check the source to verify that this is the current version before use.

3.5.2 Wheels and casters. The engine stand shall have wheels of adequate size and strength to enable the engine stand the freedom of movement a minimum distance of 25 feet when loaded to the overload capacity specified in 3.4.4. The engine stand shall have casters on two of its wheels enabling the wheel to change their direction of travel a minimum of 120 degrees when moving and loaded to the overload capacity specified in 3.4.4.

3.5.3 Immobilizing device(s). The engine stand shall incorporate a immobilizing device(s) that when utilized, shall lock or block all wheels and casters rendering the stand immobile. A locking device shall prevent all wheels from turning and casters from swirling. A blocking device shall elevate and hold the engine stands wheels off the floor while loaded to the overload capacity specified in 3.4.4.

3.5.4 Mechanical rotating mechanism. A mechanical rotating mechanism shall be provided on one of the upright posts or columns to permit one operator to manually rotate engine assemblies up to the minimum weight specified in 3.4.4. The rotating mechanism shall incorporate a crank, a pinion or worm and mating gear, or other components which shall permit easy operation. When maximum load conditions are applied, the unit shall be rotated with an applied force on the end of the crank of not greater than 30 foot-pounds. The mechanism shall function as specified in 3.4.3. The centerline of rotation of the rotating mechanism to the floor shall not exceed 40 inches (see 3.4.3). The output end of the rotating mechanism shall be a hardened steel plate no less than 10 1/2 inch diameter and 1 inch thick and shall be the mounting plate for all engine adapters. The mounting plate shall have six equally spaced 5/8 inch diameter 11 thread, N.C. tapped through holes on a 9.250 $\pm$ .005 inch diameter bolt circle for mounting of the adapter plates.

3.5.5 Mounting adapter plates and attachments. Unless otherwise specified (see 6.2), each engine stand shall be furnished with engine adapter plates and attachments for mounting the engines, listed in Table I, to the maintenance stand mounting plate. The manufacturer is encouraged to use universal type adapter plates with the capabilities of fitting as many engines as possible to keep the number of adapter plates to a minimum. The adapter plates and attachments shall be permanently and legibly marked to ensure positive identification and mating of engines and stands (see 6.2). The adapter plates shall have six equally spaced 21/32 inch diameter holes, drilled through on a 9.250 $\pm$ .005 inch diameter bolt circle for bolting the adapter plates securely to the maintenance stand mounting plate.

3.6 Tilting mechanism. A tilting mechanism shall be provided to assure clearance for large engine assemblies during the 360-degree rotation specified in 3.4.3. The mechanism shall be a hydraulic unit or a mechanical unit. The mechanism shall be capable of maintaining the 9000 minimum load capacity at heights within the upper one-third of the height range for a period of not less than 30 minutes without leakage (hydraulic unit) or a change in the height position in excess of .187 inch (mechanical unit). The height position change shall not exceed .031 inch during any five minute interval. All controls shall be accessible and so located that operating the mechanism shall not be hazardous to the operator. Means shall be incorporated into the test stand to mechanically secure the load after it has been raised in preparation for work on the engine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

3.6.1 Hydraulic unit. When the tilting mechanism is a hydraulic unit it shall incorporate a control (release) valve. The release valve shall permit the spindle, with maximum load, to return to its horizontal position with a controlled slow speed of not more than one inch per minute and shall permit stopping and holding the spindle at any position along its descent.

3.7 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean of all foreign matter detrimental to painting and have at least one coat of primer and one coat of enamel, or equivalent, applied.

3.8 Identification marking. The engine stand shall be marked for identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.9 Workmanship. Standards of workmanship shall assure that the stands shall have the strength, stability, and efficient operating characteristics found in standard commercial units and as specified in Section 3 herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4,3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified, (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of “a” through “c” below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4.4.1 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with MIL-STD-105 and the following. The size of lots for sampling for quality conformance inspection shall not be larger than 280 stands.

Examination/Test	Inspection Level	AQL
(a) Product examination (4.5)	General Level II	0.65
(b) Functional test (4.6.1)	Special Level S-4	1.0
(c) Overload test (4.6.2)	Test each stand	No failures permitted
(d) Tilting mechanism tests (4.6.3 and 4.6.3.1)	Special Level S-4	1.0
(e) Mechanical rotating mechanism test (4.6.4)	Special Level S-4	1.0
(f) Packaging inspection (4.7)	Special Level S-4	4.0

4.5 Product examination. Visually and manually examine the stand to determine conformance with the requirements of 3.2 through 3.3.7 and 3.5.1 through 3.9. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. Inspections shall be performed during the manufacturing process to the extent necessary to assure conformance; however, acceptance inspection shall be performed on completely finished (assembly) stands. Failure of the contractor to provide objective evidence that the stand has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.3.6), shall be cause for rejection.

4.6 Tests.

4.6.1 Functional test. Performance of this test necessitates the use of the largest engine assembly identified in the contract (see 6.2). The engine assembly shall be mounted on the stand in accordance with the manufacturer's instructions and shall be tested for:

(a) Movement over/against obstacles as specified in 3.4.1.
(b) Accessibility requirements specified in 3.4.2.
(c) Positioning requirements specified in 3.4.3.
(d) Locking devices' operational requirements specified in 3.5.3.
(e) Rotating mechanism operational requirements specified in 3.5.4.

Noncompliance with the requirements shall be cause for rejection of the stand being tested.

4.6.2 Overload test. A test weight of not less than 9,000 pounds shall be mounted on the engine stand and left for a period of not less than 30 minutes. At the end of the period, the stand shall be rolled a distance of not less than 100 feet to determine wheel and caster compliance with 3.5.2. The test weight shall then be removed and the stand subjected to careful examination. Any evidence of damage or deformation shall be cause for rejection of the stand (see 3.4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

4.6.3 Tilting mechanism. The engine stand shall be loaded to a minimum of 9,000 pounds. The tilting mechanism shall be actuated to raise the engine to any point within the upper one-third position of the height range. The height setting shall be recorded and the engine shall remain in this position for a period of 30 minutes. On conclusion of that time period, the height setting shall be measured to determine compliance with the requirements of 3.6.

4.6.3.1 Hydraulic unit test. The engine stand, in the level position, shall be loaded to a minimum of 9,000 pounds and then raised to the maximum height setting. The release valve shall be adjusted to permit the lowering of the spindle. The rate of descent of the spindle shall be measured, using standard measuring equipment, to determine compliance with 3.6.1, Any evidence of leakage of hydraulic fluid shall be cause for rejection of the stand being tested.

4.6.4 Mechanical rotating mechanism test. The engine stand, in the level position, shall be loaded to a minimum of 9,000 pounds. A force shall be applied on the crank to rotate the unit. This applied force shall not be greater than 30 foot-pounds, as specified in 3.5.4. Any applied force exceeding this requirement shall be cause for rejection of the stand being tested.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5 PACKAGING

5.1 Disassembly. Engine stands may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly in the field without special skills or special tools. Parts shall not be removed from the hydraulic assembly unless they can be reassembled readily in the field without special skills or special tools . Disassembly, when applicable, shall be as specified in MIL-STD-1156.

5.2 Preservation. Preservation shall be level A, or industrial, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The engine stand shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.2.1.2 Preservation application. Cranking mechanism and other operational surfaces, including the exposed surface of the ram shall be preserved with type P-7 preservative. Caster bearings shall be preserved by the application of type P-n preservative. All noncritical, unpainted or unprimed surfaces shall be coated with type P-1 preservative. The hydraulic reservoir shall be filled with hydraulic fluid and sealed to prevent leakage of the fluid. The ram shall be placed and remain in its innermost position. Cover all surfaces that have been preserved with type P-7 or type P-11 preservative with barrier material conforming to MIL-B-121, grade A, class optional and secure with a pressure sensitive tape conforming to PPP-T-60. Preservatives specified and the procedure of application shall be in accordance with MIL-P-116 and applicable documents referenced therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

5.2.1.3 Unit packaging. The engine stand shall be packaged in accordance with method I of MIL-P-116. Any technical publications shall be packaged in accordance with method IC-1 of MIL-P-116. Mounting attachments, publications, and small components removed through disassembly shall be packaged in one or more fiberboard boxes conforming to PPP-B-636, class weather-resistant. Cushioning material conforming to PPP-C-843, type II shall be used to immobilize parts within the fiberboard container and protect parts that are vulnerable to damage.

5.2.2 Industrial. Industrial preservation for each engine stand shall be in accordance with ASTM D3951-82.

5.3 Packing. Packing shall be level A, B or industrial, as specified (see 6.2).

5.3.1 Level A. Engine stand weighing 1,000 pounds or less shall be packed in a box conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Engine stands weighing over 1,000 pounds shall be individually packed in a crate, conforming to MIL-C-52950, type V, style B, covered. Anchoring, blocking, and bracing requirements shall be in accordance with MIL-STD-1186.

5.3.2 Level B. Engine stands weighing 1,000 pounds or less shall be individually packed in a box conforming to PPP-B-601, overseas type, style A, B or J. Stands weighing over 1,000 pounds shall be packed as specified in 5.3.1. Anchoring, blocking and bracing shall be as specified in 5.3.1.

5.3.3 Industrial. Industrial packing for each engine stand shall be in accordance with ASTM D3951-82.

5.4 Marking.

5.4.1 Military. When levels A or B protection are specified, military marking for interior packages and exterior shipping containers shall be in accordance with MIL-STD-129 and any special markings required by the contract.

5.4.2 Industrial. Industrial marking shall be in accordance with ASTM D3951-82.

6 NOTES

6.1 Intended use. The engine stand covered by this specification is intended for both mobile and stationary use in repair shops for the manual positioning, holding, and conveying of automotive engines, up to a maximum of 6,000 pounds, during overhaul and repair.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) First article, when required (see 3.1).
(c) Special safety devices, when required (see 3.3.3).
(d) Engine adapters/attachments, as required (see 3.5.5).
(e) Adapter/attachment identification (see 3.5.5).
(f) Painting, if different (see 3.7).
(g) Product identification marking, if different (see 3.8).
(h) First article inspection, if different (see 4.3).
(i) Quality conformance inspection, if different (see 4.4),

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

(j) Sampling, if different (see 4.4.1).
(k) Largest engine assembly required (see 4.6.1).
(l) Level of preservation, packing and marking required (see 5.2, 5.3 and 5.4).

6.2.2 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user (see 3.3.3).

6.2.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.3 Subject term (key word) listing.

AMSC number
Design
First article
First article inspection
Materials
Ordering data
Performance
Protective finish
Scope
Workmanship

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Air Force - 99

Review activities:

Air Force - 82

User activities:

Army -AT, ME

Preparing activity:

Army - AL
Project No. 4910-0578

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Armement Research and Development Command
ATTN: DRDAR-TST-S
Dover, NJ 07801

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:12Z
Check the source to verify that this is the current version before use.